

OPERATING SYSTEMS

Scheduling Algorithms

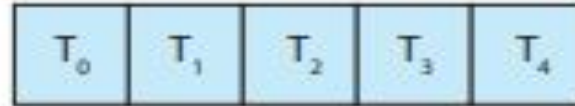
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MULTILEVEL QUEUE SCHEDULING

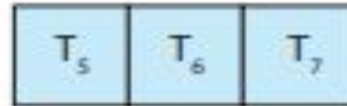
- With both priority and round-robin scheduling, all processes may be placed in a single queue, and the scheduler then selects the process with the highest priority to run.
- Depending on how the queues are managed, an $O(n)$ search may be necessary to determine the highest-priority process.
- In practice, it is often easier to have separate queues for each distinct priority, and priority scheduling simply schedules the process in the highest-priority queue.
- This approach—known as **multilevel queue**.

MULTILEVEL QUEUE SCHEDULING

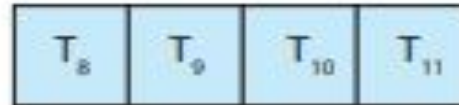
priority = 0



priority = 1



priority = 2



priority = n



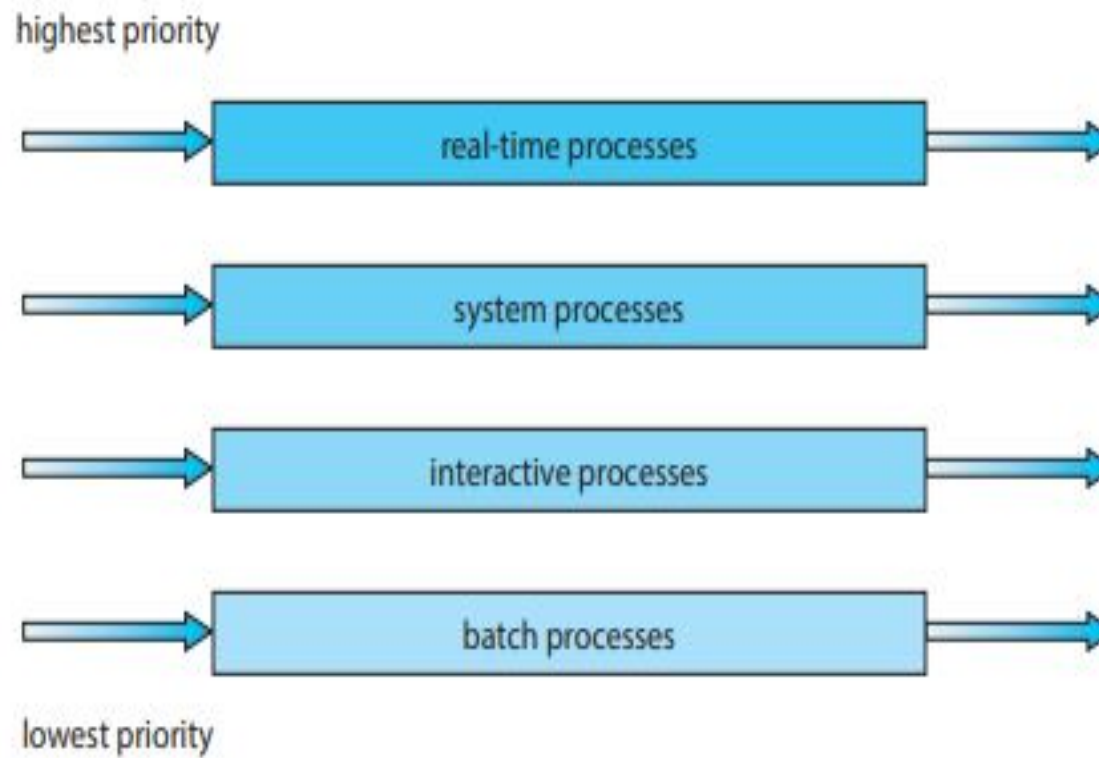
MULTILEVEL QUEUE SCHEDULING

- A multilevel queue-scheduling algorithm partitions the ready queue into several separate queues.
- Each queue has its own priority and scheduling algorithm.
- Processes are permanently assigned to one queue, generally based on some property of the process, such as memory size, process priority or process type.
- In addition, there must be scheduling among the queues, which is commonly implemented as fixed-priority preemptive scheduling i.e., serve all from the foreground then from the background.
- Another possibility is to time-slice between queues.
- Each queue gets a certain portion of the CPU time, which it can then schedule among the various processes in its queue, e.g., 80% to foreground in RR and 20% to background in FCFS.
- Scheduling across queues prevents starvation of processes in lower-priority queues.

MULTILEVEL QUEUE SCHEDULING

- A common division is made between foreground (interactive) processes and background (batch) processes.
- These two types of processes have different response-time requirements and so may have different scheduling needs.
- In addition, foreground processes may have priority (externally defined) over background processes.
- Separate queues might be used for foreground and background processes, and each queue might have its own scheduling algorithm.
- The foreground queue might be scheduled by an RR algorithm, for example, while the background queue is scheduled by an FCFS algorithm.

MULTILEVEL QUEUE SCHEDULING



MULTILEVEL FEEDBACK QUEUE SCHEDULING

- It allows a process to move between queues.
- The idea is to separate processes with different CPU burst characteristics.
- If a process uses too much CPU time, it will be moved to a lower-priority queue.
- This scheme leaves I/O bound and interactive processes in the higher-priority queues.
- Similarly, a process that waits too long in a lower-priority queue may be moved to a higher priority queue.
- This form of aging prevents starvation.

MULTILEVEL FEEDBACK QUEUE SCHEDULING

In general, a multi-level feedback queue scheduler is defined by the following parameters:

- Number of queues
- Scheduling algorithm for each queue
- Method used to determine when to upgrade a process to the higher priority queue
- Method used to determine when to demote a process
- Method used to determine which queue a process enters when it needs service

MULTILEVEL FEEDBACK QUEUE SCHEDULING

